EE 172 – UNIT 2

DC Machines

Short Detailed Notes

2.1 INTRODUCTION

What is a DC Machine?

- Essentially an **AC machine** with a **commutator**
- The commutator **converts AC to DC** (or DC to AC)
- Can operate as either a **motor** or **generator**

Classification of DC Machines

Separately Excited	Self-Excited
Field winding connected to external source	Field winding powered by its own armature
More expensive	More common
Quicker response to control	Three types: Shunt, Series, Compound

Self-Excited Types

1. **Shunt Machine** – Field winding in **parallel** with armature
2. **Series Machine** – Field winding in **series** with armature
3. **Compound Machine** – Combination of shunt and series (not covered)

2.2 SEPARATELY EXCITED MACHINES

Key Feature

- Field winding gets power from an **independent external source**
- Armature and field circuits are **electrically separate**

Applications

- Where **precise control** is needed
- Laboratory/research experiments
- Quick response to field changes

2.3 SELF-EXCITED MACHINES

Key Feature

- Field winding gets power from the **machine's own armature**
- Three types: **Shunt, Series, Compound**
- Different modes of connecting field coils to armature

2.4 DC GENERATORS

Constructional Features

Main Parts:

1. **Field Yoke** – Carries magnetic flux, provides mechanical support
2. **Pole Cores** – Support field windings
3. **Air Gap** – Between poles and armature
4. **Armature Core** – Rotating part with windings
5. **Commutator** – Converts AC to DC
6. **Brushes** – Collect current from commutator

Additional Features:

- **Interpoles (Commutating Poles)** – Placed midway between main poles
- Purpose: **Improve commutation**
- Active only when armature current flows

Magnetic Circuit: Field yoke → pole cores → air gap → armature core

Electric Circuits: Armature winding, commutator, brushes, field windings

2.4.1 Voltage Equation of DC Generator

Definitions:

Symbol	Meaning
P	Number of poles
Φ	Useful flux per pole (Webers)
Z	Total conductors on armature
N	Speed (rpm)
b	Number of parallel paths in armature
E_g	Generated emf

Formula:

$$E_g = (P \times \Phi \times Z \times N) / (60 \times b)$$

For Different Windings:

Lap Winding	Wave Winding
b = P	b = 2
Used for high current	Used for high voltage

Simplified Form:

$$E_g = K_1 \times \Phi \times N$$

where K_1 = PZ/(60b) (constant for a machine)

Example 1 – Generator Voltage Calculation

Problem: 60kW, 4-pole generator, lap winding, 48 slots × 6 conductors/slot, Φ = 0.08 Wb, N = 1040 rpm

```
Z = 48 × 6 = 288 conductors
b = P = 4 (lap winding)

E_g = (4 × 0.08 × 288 × 1040) / (60 × 4)
E_g = 400 V

Load Current: I = P/V = 60,000/400 = 150 A
Conductor current = 150/4 = 37.5 A (4 parallel paths)
```

2.4.2 Armature Reaction in Generators

Definition: The distorting effect of armature current flux on the main field flux

Causes:

- When armature delivers current, it produces its own magnetic field
- This field **interacts** with the main field

Two Main Effects:

1. **Demagnetizing** – Weakens main flux → reduces voltage
2. **Cross-magnetizing** – Distorts flux distribution → causes sparking at brushes

2.4.3 Compensating Windings

Purpose: Neutralize cross-magnetizing effect

Construction:

- Embedded in **pole faces**
- Connected in **series** with armature
- Carry **same current** as armature

Applications:

- High speed DC machines
- High voltage DC machines
- Large capacity machines

- Machines subject to large load fluctuations

2.4.4 Load Characteristics of DC Generators

Three Important Characteristics:

Characteristic	Shows Relationship
No-Load Saturation (OCC)	E_0 vs I_f (generated emf vs field current)
Internal Characteristic	E vs I_a (generated emf vs armature current)
External (Load) Characteristic	V vs I (terminal voltage vs load current)

2.4.4.1 No-Load Characteristics (Separately Excited)

- Also called **Open-Circuit Characteristic (OCC)**
- Shows relationship between **no-load generated emf** and **field current**
- Taken at **constant speed**

2.4.4.2 Load Characteristics (Separately Excited)

- Also called **Voltage Regulation Curve**
- Shows relationship between **terminal voltage** and **load current**

Why Terminal Voltage Drops Under Load:

1. **Armature reaction** – Distorts/weakens main flux
2. **IR drop** – Resistance of armature and brushes

Terminal Voltage Equation:

```
V = E_g - I_a × R_a
```

Example 2 – Voltage Change with Speed and Field

Problem: E_g = 200V at N=1200 rpm, I_f=2.5A

(a) At N=1600 rpm, I_f=2.5A:

$$E_{g2} = E_{g1} \times (N_2/N_1) \times (I_{f_2}/I_{f_1})$$

```
E_g2 = 200 × (1600/1200) × (2.5/2.5) = 266.67 V
```

(b) At N=1000 rpm, I_f=3.0A:

```
E_g2 = 200 × (1000/1200) × (3.0/2.5) = 200 V
```

Key Formula:

```
E_g ∝ I_f × N
```

2.4.5 Applications of Separately Excited Generators

- **More expensive** than self-excited
- Used when **quick response** is needed

- **Precise control** of output
- **Laboratory/research** applications

2.4.6 Shunt Generators

Connection: Field winding in **parallel** with armature

Key Equations:

```
I_sh = V / R_sh (Field current)
I_a = I_sh + I (Armature current)
V = E_g - I_a × R_a (Terminal voltage)
Power Developed = E_g × I_a
Power Delivered = V × I
```

Factors for Voltage Drop:

1. **Armature reaction** – weakens flux
2. **Armature resistance drop** – I_aR_a loss
3. **Reduction in field current** – lower voltage means lower field current

Characteristics:

- **Fairly constant** output voltage
- Rated as **constant voltage** generator

Applications:

- Supplying excitation to AC generators
- Short distance applications
- **Charging storage batteries**

Example 3 – Shunt Generator Calculations

Problem: Delivers 96A at 240V, $R_a = 0.15\Omega$, $R_f = 60\Omega$, brush drop = 2V

```
(a) Field current: I_f = 240/60 = 4A
Armature current: I_a = I_f + I_L = 4 + 96 = 100A

(b) Generated voltage:
E_g = V_t + I_aR_a + V_brush
E_g = 240 + (100 × 0.15) + 2 = 257V
```

2.4.7 Series Generators

Connection: Armature, field, and load all in **series**

Key Equations:

```
I_a = I_se = I (Same current everywhere)
V = E_g - I(R_a + R_se) (Terminal voltage)
Power Developed = E_g × I
Power Delivered = V × I
```

Characteristics:

- **Rising voltage** with load initially
- Voltage increases until **saturation**
- **Unstable** characteristic

Why Curve is Lower than Saturation Curve:

1. Armature reaction reduces flux
2. Resistance drops in armature, field, and brushes

Applications:

- **Voltage boosting** (where voltage needs to increase with current)
- Supplying field current for **regenerative braking** of DC locomotives

Limitation: Unstable characteristic limits general use

2.5 DC MOTORS

Classification

1. **DC Motors** – Run on DC supply
2. **AC Motors** – Run on AC supply
3. **Universal Motors** – Run on either AC or DC

Note: AC motors are more common because:

- Most supply systems are AC
- AC motors are **simpler and cheaper**

2.5.1 Back EMF in DC Motors

What is Back EMF?

- When motor runs, armature conductors **cut magnetic flux**
- This induces an emf (generator effect)
- Direction is **opposite** to applied voltage
- Called **counter or back emf (E_b)**

Key Points:

- E_b ∝ speed (N) and flux (Φ)
- **Same formula** as generator EMF:

```
E_b = (P × Φ × Z × N) / (60 × b) = K₁ × Φ × N
```

Armature Current:

```
I_a = (V - E_b) / R_a
```

2.5.2 Significance of Back EMF

Acts Like a Governor:

- **High speed** → E_b large → I_a small → less torque
- **Low speed** → E_b small → I_a large → more torque
- Motor is **self-regulating**

"Back EMF makes motor draw just enough current"

2.5.3 Voltage Equation of DC Motor

Fundamental Equation:

```
V = E_b + I_a × R_a
```

What This Means: Applied voltage (V) overcomes:

1. Back EMF (E_b)
2. Armature resistance drop (I_aR_a)

Power Equation (multiply by I_a):

```
V × I_a = E_b × I_a + I_a² × R_a
```

- VI_a = Power supplied to armature
- $I_a^2R_a$ = Copper loss (heat)
- E_bI_a = Mechanical power developed

Example 4 – Generator vs Motor EMF

Problem: $R_a = 0.1\Omega$, $V = 250V$

(a) As Generator ($I_a = 80A$):

```
E_g = V + I_aR_a = 250 + (80 × 0.1) = 258V
```

(b) As Motor ($I_a = 60A$):

```
E_g = V - I_aR_a = 250 - (60 × 0.1) = 244V
```

Key Difference: Generator: $E_g = V + I_aR_a$ | Motor: $E_b = V - I_aR_a$

2.5.4 Power and Torque Equations

Power Relations:

```
Input Power = V × I_a
Copper Loss = I_a² × R_a
Mechanical Power = E_b × I_a
Efficiency = E_b / V (approx.)
```

Torque Equation:

```
T_a = E_b × I_a / (2π × N)
```

After Substitution:

```
T_a = (1/2π) × Φ × Z × I_a × (P/b)
```

Simplified:

```
T_a = K₂ × Φ × I_a
```

Important: Torque is **independent of speed**!

Summary:

Generator	Motor
$E_g = K_1\Phi N$	$T_a = K_2\Phi I_a$
Voltage depends on speed	Torque depends on current

2.5.5 Speed of DC Motor

Speed Equation:

```
N = (V - I_a × R_a) / (K₁ × Φ) = E_b / (K₁ × Φ)
```

Approximation:

$$N \approx V / (K_1 \times \Phi)$$

(Since armature drop is < 5% of V)

Key Concept: Speed is **inversely proportional** to flux

Example 5 – Speed Change with Voltage and Flux

Problem: Motor runs at 900 rpm on 460V. New voltage = 200V, new flux = 0.7 × original

$$N_2/N_1 = (V_2/V_1) \times (\Phi_1/\Phi_2)$$
$$N_2 = (200/460) \times (\Phi_1/0.7\Phi_1) \times 900$$
$$N_2 = 559 \text{ rpm}$$

2.5.6 Shunt Motors

Connection:

- Field winding in **parallel** with armature
- Field rheostat in series with field for speed control

Current Relations:

```
I = I_a + I_sh (Motor)
```

(Compare to generator: $I_a = I + I_{sh}$)

Speed Characteristics

- **Practically constant speed**
- Speed drops **slightly** with load
- Classed as **constant-speed motor**

Speed Equation:

$$N = (V - I_a R_a) / (K_1 \Phi)$$

Torque Characteristics

Flux is **practically constant**

$$T_a \propto I_a$$

Torque increases **linearly** with current

Speed-Torque Characteristics

- Torque: Straight-line relationship with I_a
- Speed: Slight drop with load
- Armature reaction causes some speed increase (counteracts drop)

Applications

- **Constant speed drives**
- Machine tools
- Blowers and fans

- Lathes and woodworking machines
- Line shafts

Advantage: Safe when load is suddenly removed (no excessive speed)

2.5.7 Series Motors

Connection:

- Field winding in **series** with armature
- Same current flows through both

Current Relations:

```
I = I_a = I_se
```

(For both generator and motor)

Speed Characteristics

- Speed decreases **appreciably** with load
- Speed is **inversely proportional** to flux
- Stronger flux → lower speed

Speed Equation:

$$N = (V - I_a(R_a + R_{se})) / (K_1\Phi)$$

Torque Characteristics

Before saturation: $\Phi \propto I_a$

$$T_a \propto I_a^2$$

Torque increases as **square of current**

Example: Double current → 4× torque!

Speed-Torque Characteristics

- **High starting torque**
- Torque $\propto I^2$ (before saturation)
- Large speed variation with load

Applications

- **High starting torque** situations
- **Hoists and cranes**
- **Traction** (electric trains)
- Electric locomotives
- Wide load variations

Advantage for Traction:

- Huge starting torque
- Speed automatically adjusts to load

2.5.8 Speed Control of DC Motors

Speed Regulation

Definition: Ability to maintain constant speed for varying loads

```
SR = [(No-load speed - Full-load speed) / Full-load speed] × 100%
```

Three Control Methods

Method	How It Works	Effect
1. Armature Resistance Control	Variable resistor in series with armature	Increases R_a → decreases speed
2. Field Flux Control	Field rheostat in series with field	Increases R_f → decreases flux → increases speed
3. Voltage Control	Power electronics (thyristors)	Vary applied voltage → vary speed

Field Control Principle

- **Increasing** field resistance → **lower** flux → **higher** speed
- **Decreasing** field resistance → **higher** flux → **lower** speed

Example 7 – Speed Calculation

Problem: 600V shunt motor, R_a = 0.14Ω, No-load: N=800 rpm, I_a=12A, Full-load: I_a=225A

```
E_nl = 600 - (12 × 0.14) = 598.32 V
E_fl = 600 - (225 × 0.14) = 568.50 V
```

(a) Constant flux:

```
N_fl = (568.50/598.32) × 800 = 760 rpm
```

(b) Flux reduced to 95%:

```
N = (568.50/598.32) × (1/0.95) × 800 = 800 rpm
```

(c) Flux reduced to 80%:

```
N = (568.50/598.32) × (1/0.80) × 800 = 950 rpm
```

SUMMARY TABLE: SHUNT vs SERIES MOTORS

Feature	Shunt Motor	Series Motor
Connection	Field parallel to armature	Field series with armature
Speed	Nearly constant	Varies greatly with load
Speed Equation	$N = E_b/(K_1\Phi)$	$N = E_b/(K_1\Phi)$ (with R_{se} added)
Torque	$T \propto I_a$ (linear)	$T \propto I_a^2$ (before saturation)
Starting Torque	Moderate	Very high
Applications	Constant speed drives	High starting torque (cranes, trains)
Unsafe condition	No-load (runaway)	Light load (high speed)

KEY EQUATIONS SUMMARY

Generator

Equation	Meaning
$E_g = PZN/(60b)$	Generated emf
$E_g = K_1\Phi N$	Simplified form
$V = E_g - I_aR_a$	Terminal voltage (shunt)
$I_a = I + I_{sh}$	Current relation (shunt gen)
$V = E_g - I(R_a+R_{se})$	Terminal voltage (series gen)

Motor

Equation	Meaning
$E_b = PZN/(60b)$	Back emf
$V = E_b + I_aR_a$	Voltage equation
$T_a = K_2\Phi I_a$	Torque equation
$N = (V - I_aR_a)/(K_1\Phi)$	Speed equation
$\eta = E_b/V$	Efficiency (approx)